

标签页 1

提案 1: 贸易融资 (Trade Finance)

- 金融问题背景: 传统的国际贸易融资(如信用证处理)高度依赖纸质文件, 涉及银行、买方、卖方、海关和物流等多个中介。这导致验证过程极其缓慢、操作成本高昂, 且存在文件伪造的欺诈风险。
- 区块链的帮助 (解决方案构想): 利用区块链建立一个去中心化的文件共享和结算网络。将提单和信用证等关键文件数字化并上链, 确保其不可篡改。通过部署智能合约, 一旦物流数据(如GPS或港口扫描系统提供的预言机数据)确认货物已送达指定地点并符合条件, 智能合约将自动触发稳定币或CBDC的支付给供应商。这消除了人工审核的延迟, 降低了信任成本。

提案 2: 保险理赔 (Insurance - Parametric Insurance)

- 金融问题背景: 在农业保险或航班延误险等领域, 传统的理赔流程繁琐。受灾者需要提交证据, 保险公司需要派遣定损员进行人工核查, 导致理赔周期极长。此外, 人工定损缺乏透明度, 容易引发纠纷。
- 区块链的帮助 (解决方案构想): 引入基于区块链的参数型保险(Parametric Insurance)。将保险合同转化为智能合约, 并连接到可信的链下数据源(预言机, 例如国家气象局的降雨量数据或航空公司的航班动态系统)。当预先设定的参数被触发(如特定区域降雨量低于设定值导致干旱, 或航班延误超过3小时), 智能合约将自动执行, 并几乎实时地将理赔金打入投保人的数字钱包。这种模式大幅削减了运营和承保成本, 提高了理赔效率。

提案 3: 资产管理与投资 (Asset Management / Investments)

- 金融问题背景: 传统的优质商业房地产、私募股权或艺术品等另类资产具有极高的资金门槛, 导致普通散户投资者无法参与。同时, 这些资产缺乏流动性, 持有者很难在二级市场上快速变现。
- 区块链的帮助 (解决方案构想): 通过“真实世界资产代币化 (RWA Tokenization)”, 将高价值的非流动性资产的数字所有权转化为区块链上的证券型代币。区块链技术允许资产“碎片化”(Fractionalization), 让散户投资者可以用较小的金额购买资产的份额。同时, 通过去中心化交易所(DEX)的流动性池或自动做市商(AMM)机制, 投资者可以在全球范围内进行24/7的二级市场交易。此外, 租金收益或股息也可以通过智能合约按代币持有比例自动分发。

标签页 2

Stage 1: 3 Areas of Finance for Blockchain Assignment

Area 1: Trade Finance & Institutional Clearing (贸易融资与机构清算)

1. Existing problems and issues (现有问题与挑战):

- **EN:** Traditional international trade finance and cross-border clearing rely on correspondent banking networks (e.g., SWIFT) and manual documentation (like Letters of Credit). This involves multiple intermediaries, leading to slow settlement times (often T+1 to T+3), high foreign exchange spreads, trapped liquidity, and significant risks of document tampering.
- **CN:** 传统的国际贸易融资和跨境清算依赖于代理行网络(如SWIFT)和基于纸质的手工单据(如信用证)。这涉及多个中介机构,导致结算时间缓慢(通常为 T+1 到 T+3)、高昂的外汇点差、资金流动性受限,以及极大的单据篡改风险。

2. How blockchain can help (区块链如何提供帮助):

- **EN:** Blockchain provides a shared, immutable ledger that acts as a single source of truth. By using wholesale Central Bank Digital Currencies (CBDCs) or commercial bank digital money, financial institutions can achieve atomic, real-time cross-border settlement. Smart contracts can also automate conditional payments once digitized trade documents (like Bills of Lading) are verified.
- **CN:** 区块链提供了一个共享的、不可篡改的账本,作为单一的事实来源。通过使用批发型央行数字货币(wCBDC)或商业银行数字货币,金融机构可以实现原子化的实时跨境结算。一旦数字化的贸易单据(如提单)被验证,智能合约还可以自动执行条件支付。

3. Real-World Case Study (真实世界案例):

- **EN: Partior.** It is a blockchain-based interbank clearing and settlement network jointly established by DBS Bank, J.P. Morgan, and Temasek. It enables banks to settle cross-border transactions in different currencies in real-time using commercial bank digital money, significantly reducing settlement delays.
- **CN: Partior**。这是一个由星展银行、摩根大通和淡马锡联合成立的基于区块链的银行间清算和结算网络。它使银行能够使用商业银行数字货币实时结算不同货币的跨境交易,从而大幅减少结算延迟。

Area 2: Parametric Insurance (参数型保险)

1. Existing problems and issues (现有问题与挑战):

- **EN:** The traditional insurance claims process is notoriously cumbersome and opaque. In sectors like agriculture or travel, policyholders must manually submit evidence of loss, which is followed by time-consuming manual assessments by adjusters. This results in delayed payouts, high administrative costs for the insurer, and frequent disputes over payout amounts.
- **CN:** 传统的保险理赔流程极其繁琐且不透明。在农业或旅游等领域,投保人必须手动提交损失证据,随后由理赔员进行耗时的人工评估。这导致赔付延迟、保险公司的管理成本高昂,以及频繁的理赔金额纠纷。

2. How blockchain can help (区块链如何提供帮助):

- **EN:** Insurance policies can be coded as smart contracts linked to trusted off-chain data sources (Oracles). For instance, if an oracle feeds data showing a flight is delayed by >3 hours, or a region's rainfall drops below a threshold, the smart contract executes automatically. This provides near-instant payouts directly to the user's wallet without human intervention.
- **CN:** 保险单可以被编写为链接到可信链下数据源(预言机)的智能合约。例如,如果预言机输入的数据显示航班延误超过3小时,或者某地区的降雨量低于阈值,智能合约将自动执行。这可以直接向用户的钱包提供近乎实时的赔付,无需人工干预。

3. Real-World Case Study (真实世界案例):

- **EN: Etherisc.** It is a decentralized insurance protocol that has successfully deployed parametric crop insurance in Africa. Using Chainlink oracles to fetch localized weather data, the smart contracts automatically trigger payouts to farmers' mobile wallets during severe droughts, bypassing traditional claims adjusters entirely.
- **CN: Etherisc.** 这是一个去中心化保险协议,已在非洲成功部署了参数型农作物保险。它使用 Chainlink 预言机获取当地天气数据,在严重干旱期间,智能合约会自动触发向农民移动钱包的赔付,完全绕过了传统的理赔员。

Area 3: Asset Management & Alternative Investments (资产管理与另类投资) - [Target Topic]

1. Existing problems and issues (现有问题与挑战):

- **EN:** Premium alternative assets (such as private equity, commercial real estate, or high-yield fixed income) have extremely high capital barriers to entry, making them inaccessible to retail investors. Additionally, these markets lack secondary liquidity; ownership transfers are slow, legally complex, and require expensive intermediaries.
- **CN:** 优质的另类资产(如私募股权、商业房地产或高收益固定收益产品)具有极高的资金准入门槛,使得散户投资者无法触及。此外,这些市场缺乏二级流动性;所有权转移缓慢、法律程序复杂,且需要昂贵的中介机构。

2. How blockchain can help (区块链如何提供帮助):

- **EN:** Through Real-World Asset (RWA) tokenization, the digital ownership of high-value assets can be fractionalized, democratizing access for smaller investors. The use of decentralized networks and Automated Market Makers (AMMs) facilitates 24/7 global secondary market trading, unlocking deep liquidity. Smart contracts can also streamline asset servicing, such as automated coupon and redemption payments.
- **CN:** 通过真实世界资产(RWA)代币化,高价值资产的数字所有权可以被“碎片化”,从而实现中小投资者的投资民主化。去中心化网络和自动做市商(AMM)的使用促进了全球全天候的二级市场交易,释放了深度流动性。智能合约还可以简化资产服务,例如自动化的息票和赎回支付。

3. Real-World Case Study (真实世界案例):

- **EN: Project Guardian (MAS) & Schrodgers.** Initiated by the Monetary Authority of Singapore, this project involves major institutions like Schrodgers, Citi, and Standard Chartered exploring tokenized investment vehicles. For example, Schrodgers and Calastone have explored tokenizing mutual funds (Variable Capital Companies - VCC) on a public blockchain, allowing for atomic settlement and automated fund management via the Guardian Funds Framework.
- **CN: Project Guardian** (新加坡金管局) 与施罗德集团。由新加坡金管局发起，该项目汇集了施罗德、花旗和渣打等大型机构，共同探索代币化投资工具。例如，施罗德和Calastone探索了在公共区块链上将共同基金(可变资本公司 - VCC)代币化，通过Guardian Funds Framework (Guardian基金框架)实现原子结算和自动化基金管理。

Reference / Sources (供您参考和添加引用)

- **Partior:** Official press releases from MAS (Monetary Authority of Singapore) and Partior's official documentation. Mentioned in SC6125 Lecture 1 slides (Project Ubin & Cross Border Payments).
- **Etherisc:** World Economic Forum (WEF) reports on blockchain in insurance, and Etherisc's official whitepapers on decentralized crop insurance.
- **Project Guardian:** MAS official press releases regarding "Project Guardian" and the "Guardian Funds Framework (GFF)". Mentioned explicitly in SC6125 Lecture 1 slides as an ongoing initiative to test asset tokenisation.

标签页 3

Stage 2 提交方案

在Padlet的 Stage 1 和 Stage 2 之间的 + 号处点击添加, 发布以下内容:

Title: Stage 2 Decision - Group 3: Scaling Maritime Resilience via Blockchain

Body: We have decided to proceed with **Area 2: Parametric Insurance**, specifically focusing on **Maritime Shipping & Supply Chain Parametric Insurance for Perishable Goods**.

Following Dr. Teo's feedback, we are moving beyond generic models to address the "trust gap" in global logistics. Our solution targets SME importers in Singapore whose high-value, time-sensitive cargo (e.g., pharmaceuticals, fresh produce) is vulnerable to port congestion and climate volatility.

The core value proposition of our blockchain-based approach includes:

- **Trustless Automation:** Replacing subjective, manual claims adjustments with **Smart Contracts** that execute automatically based on objective, verifiable triggers.
- **Decentralized Oracle Integration:** Utilizing **AIS (Automatic Identification System)** and **IoT sensor data** as decentralized oracles to provide real-time, tamper-proof evidence of delays or temperature excursions.
- **Instant Liquidity:** Enabling **atomic payouts** in stablecoins the moment a parametric threshold is hit, providing SMEs with immediate financial buffers—a feat impossible under traditional insurance cycles.

By leveraging Singapore's status as a premier maritime hub, we aim to demonstrate how blockchain can transform supply chain risk from a slow, opaque process into a transparent, programmable financial instrument.

Stage 3 提交方案 (脑力激荡 5 个区块链解决方案)

在Padlet的 Stage 2 和 Stage 3 之间的 + 号处点击添加, 发布以下内容。这五个 Idea 从不同维度(预言机触发、DeFi资金池、ESG结合)切入了海运参数保险, 供老师在 Stage 4 之前评估。

Title: Stage 3: 5 Blockchain Ideas for Maritime Parametric Insurance

Body: Following our decision to focus on **Parametric Insurance for Maritime Shipping**, our group brainstormed 5 distinct blockchain-based solutions targeting different financial pain points in the industry:

Idea 1: Oracle-Driven Port Delay Payouts for SME Importers

- **Problem:** SMEs importing perishable goods face inventory loss and cash flow crunches when vessels are delayed by port congestion. Traditional claims take months.
- **Blockchain Solution:** A smart contract integrated with maritime AIS (Automatic Identification System) data oracles. If the oracle verifies a specific vessel is stuck at anchorage for >72 hours, the contract automatically executes a stablecoin payout to the SME, providing instant liquidity to re-order goods or cover operational costs.

Idea 2: Decentralized Underwriting Liquidity Pools (DeFi Integration)

- **Problem:** Traditional shipping insurance requires massive capital reserves from centralized insurers, leading to high premiums and limited coverage options for niche routes.
- **Blockchain Solution:** Creating a peer-to-peer (P2P) DeFi liquidity pool where institutional/retail investors stake stablecoins to underwrite specific maritime risks (e.g., "Q3 Malacca Strait Weather Risk"). Investors earn premiums as yield. If a parametric trigger is hit, the smart contract automatically withdraws from the pool to pay the policyholder, ensuring transparent and fully collateralized underwriting.

Idea 3: ESG & Carbon Offset Parametric Triggers for Route Deviations

- **Financial Problem:** When cargo vessels encounter extreme weather (e.g., cyclones), they are forced to make significant route deviations. This results in the burning of excess bunker fuel, causing shipping companies to breach strict corporate ESG covenants and face substantial financial penalties (such as carbon taxes or higher interest rates on green loans).
- **Blockchain Solution:** A parametric insurance smart contract linked to both global weather APIs and maritime AIS (routing) oracles. If a verified detour occurs due to severe weather, the contract automatically calculates the excess emissions. Instead of a traditional fiat payout, the smart contract executes an atomic transaction using stablecoins to instantly purchase and retire tokenized Carbon Credits on a decentralized registry (aligning with MAS's Project Greenprint). This provides real-time, programmable ESG compliance and directly mitigates the carbon penalty risk.

Idea 4: IoT-Linked Cold Chain Spoilage Insurance

- **Problem:** High-value cold chain logistics (pharmaceuticals, premium seafood) often suffer temperature excursions. Traditional claims involve lengthy disputes over *when* and *where* the goods spoiled.
- **Blockchain Solution:** Integrating IoT temperature sensors inside refrigerated containers (reefers) as oracles. If the internal temperature exceeds a predefined threshold for more than 2 hours, the smart contract registers the breach on-chain and triggers an immediate payout to the cargo owner, eliminating forensic investigations and dispute resolution costs.

Idea 5: "Container Rollover" Business Interruption Insurance

- **Problem:** SMEs frequently have their containers "rolled" (bumped to the next available ship) by major carriers to favor larger clients, causing severe business interruption and missed delivery penalties.
- **Blockchain Solution:** A micro-insurance smart contract linked to port terminal API data. If the oracle confirms the specific container ID failed to board the scheduled vessel before departure, the contract instantly transfers a predetermined stablecoin payout to the SME to cover emergency airfreight costs or client compensation.

标签页 4

Good evening everyone. We are Group 3, and my name is Zhang He.

Our redesign idea is simple: **we redesign Terra by adding rate-limited redemptions and state-dependent discount curves, so the system can slow down exits and share losses during panic, instead of triggering unlimited LUNA minting and a death spiral.**

To briefly recap, Terra's original mechanism allowed **1 UST to be redeemed for 1 dollar worth of LUNA**. This worked in normal times, but under a panic, large-scale redemptions could force the protocol to mint too much LUNA and push the system into a death spiral.

So our main goal is not to defend the 1-dollar peg at any cost. Instead, our goal is to answer one question: **when many users rush to exit, how can we stop unlimited LUNA minting and keep the system alive?**

Now let's look at our workflow.

Before going into the flow, I want to introduce the two key mechanisms in our redesign.

The first mechanism is **rate-limited redemption**. This means the protocol sets a daily cap on how much UST can be converted into LUNA. So even if many users want to exit at the same time, the system will not allow unlimited redemption in one day.

The second mechanism is the **state-dependent discount curve**, or haircut. This means the redemption value changes according to the stress level of the system. In normal times, 1 UST can still be redeemed for 1 dollar worth of LUNA. But when the system is under stress and the redemption queue becomes longer, the redemption value gradually decreases. For example, 1 UST may only be redeemed for 90 cents or 80 cents worth of LUNA.

With these two mechanisms, the flow becomes much safer.

In the **normal state**, the system still follows Terra's original logic: 1 UST can be redeemed for 1 dollar worth of LUNA, so the basic arbitrage mechanism is preserved.

In the **stress state**, when a demand shock happens, the redemption queue starts to grow. Then the daily cap slows down redemptions, and the haircut reduces the amount of LUNA minted for each UST. As a result, LUNA inflation is limited, selling pressure is reduced, and the death spiral is controlled.

So, this is our basic workflow: we slow down exits, reduce the amount of LUNA minted, and contain the death spiral. Next, my teammate Huang Ziteng will explain the main trade-off behind this design and why we think this trade-off is necessary.

Finally, let's talk about the main trade-off.

Our trade-off is very clear: **we give up a strict 1-dollar peg and instant liquidity in exchange for system survivability.**

In normal times, this cost is small, because redemption can still work close to the original mechanism. But in crisis times, users may not be able to exit immediately, and they may receive less than 1 dollar in value.

However, we think this trade-off is necessary. Without enough external reserves, defending a perfect 1-dollar peg during a massive panic means the protocol has to keep minting LUNA endlessly. That may protect exiting users in the short term, but it can destroy the whole ecosystem.

So our design chooses a more realistic path: **normal redemption in normal times, controlled loss-sharing in crisis times, and survival of the ecosystem as the top priority.**

标签页 5

First, **StableSwap**, represented by Curve, is mainly designed for assets with very similar values, such as USDC, USDT, and DAI. Its core idea is to use a **hybrid AMM function**. When the pool is balanced and the assets trade near the peg, the curve behaves more like a **constant sum function**, so the price curve is very flat. This allows large stablecoin swaps with very low slippage. But when the pool becomes seriously imbalanced, the function gradually behaves more like a **constant product function**, which prevents the pool from being drained. So StableSwap reduces slippage by designing a flatter curve around the expected price range.

Second, **Uniswap v3** reduces slippage through **concentrated liquidity**. Unlike traditional AMMs where liquidity is spread across the whole price range, Uniswap v3 allows liquidity providers to place their funds only within selected price ranges. This makes liquidity much deeper around the current market price. As a result, trades inside that active range face lower slippage because there is more effective liquidity available.

So the key difference is: **StableSwap reduces slippage by changing the pricing curve for stable or highly correlated assets**, while **Uniswap v3 reduces slippage by concentrating liquidity in the most useful price range**. StableSwap is especially suitable for stablecoin swaps, whereas Uniswap v3 is more flexible and can be used for many trading pairs, but it requires more active liquidity management.

标签页 6

SC6125 Group 3 Task Distribution

Project Title: Scaling Maritime Resilience via Blockchain: Parametric Insurance for SME Importers (OceanNode)

Target Group: SME importers in Singapore (Pharmaceuticals, fresh produce)

1. Core Task Allocation (核心任务分配)

Team Member	SC6125 Component & Focus Area	Specific Responsibilities (Based on Course Guidelines)
Jin Jiayu	1. Problem Definition	Finance First Principle: Clearly define the financial problem (SME trapped liquidity due to delays) and economically motivate it .
	2. System Architecture	• Identify users and stakeholders with clear incentives . • Provide strong evidence on why existing manual insurance claims are inadequate . • Design a high-level system architecture showing clear trust boundaries, on-chain vs. off-chain components, and participant roles.

		this is not a full report) ensuring content is well-structured and logical.
--	--	---

2. Key Coordination Notes (协作要点)

- **Integration over Isolation:** Although He Zhili, HUANG ZITENG, and LI JIAWEI are handling Ideas 1, 2, and 5 respectively, the final presentation must present these as **one coherent business model** rather than three disjointed products. Jin Jiayu's architecture diagram must visually unify all three ideas.
- **No Functional Code Required:** For the Smart Contract / dApp Design (Component 3), focus purely on pseudocode, solidity style structures, or sequence diagrams. The professor explicitly stated there is no requirement for deployable or fully functional code.
- **Presentation Flow (For He Zhili):** Ensure the slide deck follows the suggested flow: Financial problem $\rightarrow$ Proposed solution $\rightarrow$ System logic $\rightarrow$ Digital asset $\rightarrow$ Key risks.
- **Summary Alignment (For Zhang He):** The 2-3 page written summary must accurately reflect the final slide deck. Since it accounts for 10% of the grade, ensure the poster/summary is self-explanatory and any external visuals used are properly cited.

Summary

1. Problem Definition

2. System Architecture

3. Ideas and Core Design

OceanNode integrates three ideas into one coherent Maritime Resilience Policy rather than treating them as separate insurance products. Idea 1 protects SMEs against post-departure port congestion delays. Idea 5 protects them against pre-departure container roll-overs. Idea 2 provides the decentralized underwriting liquidity pool that funds the payouts for both risk events. Together, these modules create a full-stack blockchain-enabled insurance model: SMEs buy coverage, external oracles verify real shipping events, smart contracts execute payouts, and liquidity providers supply underwriting capital.

3.1 Idea 1: Port Delays

The first risk module in OceanNode addresses post-departure port congestion delays. This problem is especially important for SME importers in Singapore that handle time-sensitive cargo such as pharmaceuticals, fresh produce, and premium food products. When a vessel is forced to wait at anchorage near the destination port for several days, the importer may suffer cargo spoilage, missed delivery commitments, penalty costs, and a serious working-capital gap. Traditional marine insurance is not well suited to this type of loss because the claim process usually requires manual documentation, surveyor reports, proof of financial loss, and insurer assessment. For small importers, the payout may arrive weeks or months after the liquidity shock has already occurred.

OceanNode turns this delay risk into a parametric insurance trigger. Under the unified Maritime Resilience Policy, the SME submits shipment details such as the vessel IMO number, destination port, cargo category, coverage amount, and wallet address when purchasing coverage. After the vessel departs, the Policy Manager Smart Contract monitors the shipment through AIS maritime oracle data. If the AIS oracle verifies that the vessel remains within the defined anchorage zone of the destination port for more than 72 consecutive hours, the port-delay condition is treated as objectively satisfied. The contract does not need to investigate the exact commercial loss of each shipment; instead, it pays a predefined amount according to the agreed policy terms.

This module creates a clear financial improvement over traditional insurance. For the SME, the benefit is immediate liquidity: the stablecoin payout can be used to reorder replacement goods, arrange emergency logistics, or cover penalties owed to downstream buyers. For liquidity providers, the same trigger defines when their underwriting capital is at risk. If no port-delay event occurs before the policy expires, the premium remains in the risk pool as yield. If the trigger is confirmed, the Risk Pool Smart Contract releases the claim payment automatically. This makes the underwriting economics transparent and rule-based.

The key reason blockchain is useful here is not simply that the process is digital, but that it combines objective data, programmable settlement, and transparent records. AIS data provides an external factual input; the smart contract converts that input into an enforceable payout decision; and the blockchain records policy status, trigger confirmation, and settlement history. Therefore, the port-delay module helps OceanNode reduce the trust gap between SMEs, capital providers, and insurers while keeping the product financially focused on cash-flow protection.

3.2 Idea 2: DeFi Pools

The second module in OceanNode provides the decentralized underwriting liquidity pool that funds the payouts for both port-delay and container roll-over events. While Idea 1 and Idea 5 define the insured risks, Idea 2 defines where the claim money comes from and how underwriting capacity is created. In traditional marine insurance, this role is usually performed by centralized insurers with their own balance sheets. However, for small, route-specific, and time-sensitive risks, traditional underwriting can be expensive, slow to scale, and unattractive for insurers.

OceanNode replaces this model with a blockchain-based risk pool. Liquidity providers deposit stablecoins, such as USDC, into a dedicated maritime risk pool. This pool acts as the capital reserve for active insurance policies. When an SME purchases a Maritime Resilience Policy, the premium is routed into the risk pool and becomes yield for liquidity providers. If no insured event occurs, LPs retain the premium as underwriting income. If a valid port-delay or roll-over trigger is confirmed, the Risk Pool Smart Contract releases the payout directly to the SME.

This design makes LPs act as decentralized underwriters. They are not simply providing passive liquidity; they are taking on clearly defined maritime risk in exchange for premium income. Their capital is exposed when policies are active, but the exposure is transparent because pool liquidity, active coverage, premium inflows, and claim payouts can be recorded on-chain. This creates a more rule-based and auditable underwriting model compared with traditional insurance.

The main financial benefit of this module is that it connects SME insurance demand with decentralized capital supply. SMEs gain access to faster and more flexible protection, while LPs gain a new source of risk-adjusted yield. However, the model also requires careful risk management. If the pool sells too much coverage or underprices the risk, LPs may suffer losses when many claims occur at the same time. Therefore, OceanNode must apply capacity limits, exposure caps, and conservative pricing rules to keep the pool solvent.

3.3 Idea 5: Roll-overs

[[LI JIAWEI to insert container roll-over section.]]

3.4 Smart Contract / dApp Design

要把idea 1, 2, 5的contract都放进来

The policy is executed through smart contracts. Instead of relying on manual claims assessment, the payout is triggered automatically when trusted external data confirms that a predefined condition has been met. The underwriting capital is provided by a decentralized liquidity pool, where liquidity providers deposit stablecoins and earn premiums as yield. In

this model, SMEs buy protection, liquidity providers supply underwriting capital, oracle networks verify real-world shipping events, and smart contracts execute payouts. Key roles include the SME importer, liquidity providers, the Risk Pool Smart Contract, the Policy Manager Smart Contract, oracle networks, and the governance layer. The SME purchases coverage and receives payout if a trigger event occurs. Liquidity providers supply stablecoin capital to the risk pool. The Risk Pool Smart Contract holds underwriting capital and releases payouts, while the Policy Manager Smart Contract issues policies and manages policy status. Oracle networks provide verified carrier, port, and AIS vessel data. Governance adjusts parameters such as risk pool scope, premium rates, and trigger thresholds.

For the port-delay module, the key smart contract condition is objective and time-based. The policy stores the vessel IMO number, destination port, anchorage radius, payout amount, delay threshold, and expiry time. Once the vessel approaches the destination port, the AIS oracle network monitors whether it remains inside the defined anchorage zone. If the vessel berths before the threshold is reached, the policy expires without payout. If the vessel remains in the anchorage zone for more than 72 consecutive hours, the Policy Manager marks the policy as triggered and requests payout from the Risk Pool Smart Contract.

改成diagram

The simplified state transition for Idea 1 is:

Quoted -> Active -> MonitoringAtDestination -> Triggered -> Paid

If no valid delay occurs, the alternative path is:

Quoted -> Active -> MonitoringAtDestination -> Expired

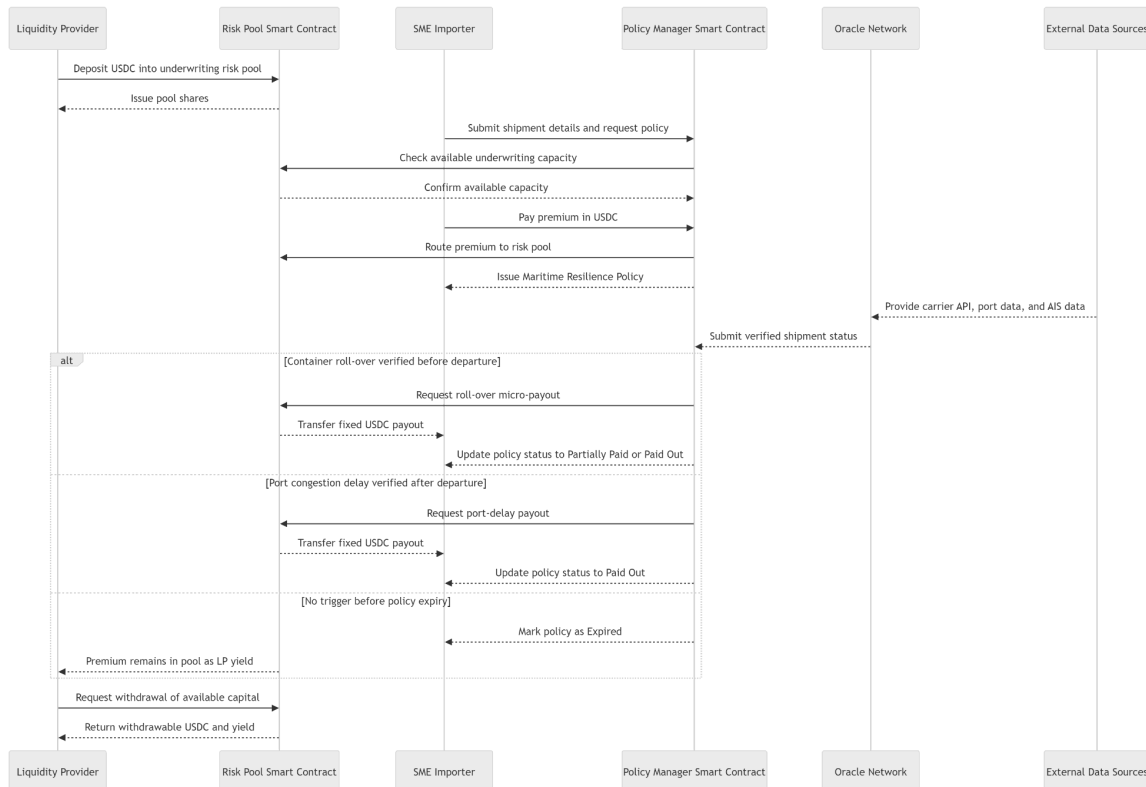
[[LI JIAWEI to insert Trigger A roll-over flow.]]

This state transition shows the financial logic of decentralized underwriting. LP capital is only freely withdrawable when it is not backing active policies. Once it is reserved, it becomes part of the insurance capacity of the system. This prevents the protocol from issuing policies without sufficient funds and helps ensure that valid claims can be paid automatically.

A simplified version is shown below:

LP Capital State	Trigger/Event	Outcome
Deposited / Available	LP stakes USDC into the pool	LP receives pool shares
Reserved	Policy is issued	Capital is locked as underwriting capacity
Premium-Earning	SME pays premium	Premium increases pool yield
Claimed	Valid trigger is verified	Pool pays SME from reserved capital
Released	Policy expires without claim	Capital becomes available again
Withdrawable	LP exits after lock period	LP withdraws available capital and yield, minus losses

Sequence Diagram



3.5 Digital Asset Comparison

OceanNode uses stablecoins, especially USDC, as the operational settlement asset for premiums and payouts. This is appropriate because the insured SMEs need a predictable claim value, not exposure to cryptocurrency price volatility. USDC also allows premiums, reserves, and claim payouts to be handled directly by smart contracts.

Compared with volatile crypto assets such as ETH, stablecoins are more suitable because the payout amount remains close to a fiat-denominated business loss. Compared with CBDCs or deposit tokens, stablecoins are more practical for a prototype because they are already widely used in DeFi applications and can be integrated with liquidity pools. However, CBDCs or tokenized bank deposits may become stronger options for a regulated institutional version of OceanNode in the future.

There are still important risks. Stablecoins depend on issuer reserves, redemption confidence, and regulatory treatment. A depeg event could reduce the real value of a payout, while issuer controls may affect transferability. For this reason, OceanNode should use reputable fully backed stablecoins, disclose issuer and depeg risks, and avoid using the optional governance token \$NODE as the payout asset. \$NODE, if used, should remain only an incentive or governance layer for LP participation.

[[LI JIAWEI to add asset comments for LP capital and roll-over micro-payouts.]]

For the Idea 2 liquidity pool, the asset choice mainly affects pool solvency, LP return calculation, and the credibility of underwriting capacity. The risk pool should use USDC as its base accounting and reserve asset because LPs need a clear view of

how much capital is deposited, how much is reserved for active policies, how much premium income has been earned, and how much capital remains after claims. Using a stable unit of account makes the pool's loss ratio, utilization ratio, and yield easier to measure.

Alternative assets create different trade-offs. CBDCs could reduce private issuer risk and improve regulatory confidence, but they are not yet practical for an open DeFi-style underwriting pool. Deposit tokens may be useful if OceanNode works with banks or licensed insurers, but they could limit participation to approved institutions and reduce composability. Volatile crypto assets would add market risk on top of maritime underwriting risk, making it harder to determine whether LP losses came from claim events or asset price movements.

Therefore, the recommended design is to keep the underwriting pool's core reserve in USDC, while treating other assets as future or secondary options. CBDCs and deposit tokens may be suitable for a more institutional version of OceanNode, but they should not be assumed in the prototype. The optional \$NODE token should also remain outside the core reserve. It can be used to reward early LP participation or support governance, but it should not determine pool solvency or claim-paying capacity.

4. Evaluation & Valuation

4.1 Benefits

OceanNode offers three major benefits. First, it reduces claim-processing delays by replacing manual assessment with objective oracle-based triggers. This is especially valuable for SMEs because the main financial problem is often liquidity timing rather than only final loss compensation. Second, it lowers administrative cost for small and frequent insurance events, such as container roll-overs, because the payout process is standardized and automated. Third, the DeFi liquidity pool creates an alternative source of underwriting capital, allowing maritime risk to be funded by investors who are willing to earn risk-adjusted premium yield.

4.2 Limitations and Risks

The main limitation is oracle dependency. OceanNode can only be as reliable as the off-chain data it receives from AIS feeds, carrier APIs, and port systems. If the data is delayed, incomplete, or manipulated, the contract may trigger incorrectly or fail to trigger when SMEs expect a payout. Smart contract risk is another concern because bugs in policy or pool contracts could affect customer payouts or LP funds. Stablecoin risk also remains relevant: if the settlement asset depegs or faces issuer problems, the economic value of the payout may be weakened.

[[ZHANG HE / team to add DeFi liquidity, LP loss, adverse selection, and pricing risks.]]

4.3 Adoption, Regulation, and Scalability

Adoption depends on whether SMEs, logistics platforms, insurers, and data providers are willing to connect to the same system. SMEs may value faster payouts, but they may also

need simple onboarding, fiat conversion, and assurance that the product is legally enforceable. From a regulatory perspective, OceanNode must consider insurance distribution rules, KYC/AML requirements, treatment of stablecoins or digital payment tokens, and the governance of pooled underwriting capital.

The model is scalable if it starts with narrow, clearly defined risk pools such as Singapore-bound perishable imports, then expands to other ports, routes, cargo types, and delay thresholds. The same architecture can support multiple maritime risk modules as long as the trigger conditions are objective and supported by reliable data. However, scaling should be gradual because inaccurate pricing or excessive correlated claims could drain the liquidity pool during major port disruptions.

[[ZHANG HE to add final business viability / valuation paragraph.]]